A Riemannian Geometric Reconstruction of Quantitative Social Science: Lie Derivatives, Morse Theory, and a Geometric Answer to the Lucas Critique

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Abstract

This paper proposes a systematic reformulation of the principal analytical methods of quantitative social science — econometrics, computational political science, quantitative sociology, and cliometrics — within the Riemannian geometric framework of Geometric Intelligence (GI) theory. We demonstrate that the standard toolkit of quantitative social science — vector autoregression (VAR), Granger causality, the Chow test, simultaneous equations models, the synthetic control method, NOMINATE spatial models, Bayesian structural time-series models, social network analysis, and agent-based models — admits natural and informative extensions on Riemannian manifolds constructed from data via the pullback metric of a variational autoencoder (VAE) decoder.

The central contribution is a geometric reformulation of the Lucas critique (1976). We argue that the Lucas critique, expressed in the language of Riemannian geometry, amounts to the error of employing a fixed-metric model when the metric is not a Killing field — that is, when the Lie derivative of the metric with respect to the policy vector field is non-vanishing. The Lie derivative of GI theory measures structural change in a manner that is free of structural model assumptions (utility functions, production functions, rational expectations), offering a response to the Lucas critique that operates on a different plane from that of dynamic stochastic general equilibrium (DSGE) models.

We further observe that whilst the method of constructing the manifold is discipline-dependent (from theory in physics, from data in social science), the analytical tools available on the manifold once constructed — Lie derivatives, covariant derivatives, scalar curvature, geodesics, Morse functions — are discipline-independent. This observation implies that quantitative social scientists may share a common mathematical language with physicists, chemists, and engineers for the analysis of data on manifolds.

Keywords: quantitative social science, cliometrics, Riemannian manifold, Lie derivative, Killing field, scalar curvature, Morse theory, Lucas critique, DSGE, VAR, Granger causality, Chow test, synthetic control method, variational autoencoder, pullback metric, Geometric Intelligence

JEL Classification: C10, C32, C45, C63, N01, B41

1. Introduction: The Methodological Plateau of Quantitative Social Science

1.1 The problem

Cliometrics — the application of quantitative methods to historical research — was established by Fogel (1964) and North (1990), who introduced econometric techniques into the discipline of history and were

jointly awarded the Nobel Memorial Prize in Economic Sciences in 1993. In the six decades since Fogel's celebrated counterfactual analysis of American railroads, the methodological foundations of cliometrics have remained, in their essentials, unchanged. The toolkit consists of linear regression, vector autoregression (VAR), difference-in-differences (DID), instrumental variables (IV), and cognate techniques drawn from standard econometrics. The journals of the field — Explorations in Economic History, The Journal of Economic History, Cliometrica — continue to publish work employing these methods, and advances have been incremental rather than foundational.

An analogous plateau is discernible in computational political science. The NOMINATE model of Poole and Rosenthal (1985, 1997), which estimates the ideological positions of legislators from roll-call voting patterns, is in essence a principal component analysis (PCA) — a linear projection into a flat (Euclidean) space. In quantitative sociology, network analysis in the tradition of Granovetter (1973) operates on discrete graph structures and has not been extended to continuous geometry.

1.2 Purpose of this paper

This paper proposes that the Riemannian geometric framework of Geometric Intelligence (GI) theory (Etale Cohomology, 2026, Vols. 1–2) provides a principled and informative extension of the standard quantitative social science toolkit. We establish correspondences between ten existing methods and their Riemannian counterparts, and argue that these correspondences reveal structural information that the existing methods, by construction, cannot capture.

The correspondences are summarised in the following table.

Existing method	GI-theoretic counterpart	Core of the extension
VAR (time lags)	Lie-derivative flow	Nonlinear lag structure
Granger causality	Lie bracket [X, Y]	Non-commutativity as structural causality
Chow test	Scalar curvature discontinuity	Shape, not merely presence, of break
SEM	Multiple vector fields on the manifold	Nonlinear simultaneous determination
Lucas critique	Non-Killing metric under policy vector field	Model-free measurement of structural change
Synthetic control	Counterfactual manifold construction	Counterfactual without control group
NOMINATE	Nonlinear VAE latent space	Curved ideological space
BSTS / CausalImpact	Bayesian credible intervals on Lie derivative	Uncertainty of structural transformation depth
Network analysis	Manifold embedding	Discrete to continuous
ABM	Vector fields on manifold	Agents on curved space

2. Preliminaries: The GI Theory Pipeline for Manifold Construction

The GI theory pipeline constructs a Riemannian manifold from social science data in the following stages.

Stage 1: Data preparation. One assembles a multivariate time series $x(t)$ in R^D, where D denotes the number of variables and t indexes time.

Stage 2: Dimensionality reduction via a variational autoencoder (VAE). The D-dimensional data are compressed to a d-dimensional latent space ($d \ll D$). The encoder $q_\phi(z|x)$ maps data to latent variables z; the decoder $p_\theta(x|z)$ reconstructs the data from the latent variables.

Stage 3: Definition of the pullback metric. The Jacobian $J_f(z)$ of the decoder $f: R^d \to R^D$ induces a pullback metric on the latent space:

$$g_{ij}(z) = [J_f(z)^T J_f(z)]_{ij}$$

This metric endows the latent space with the structure of a Riemannian manifold (Z, g). The metric defines, at each point, a rule for measuring distances — a rule that varies from point to point. This is the essence of Riemannian geometry: the properties of the space are position-dependent.

Stage 4: Differential-geometric computation on the manifold. Scalar curvature R(z), the Lie derivative $L_X g$, geodesics, Morse-function critical points, and related quantities are computed.

The crucial observation is that whilst Stages 1–3 are discipline-dependent, Stage 4 is discipline-independent. In physics, the manifold is derived from theory (the spacetime of general relativity, the configuration space of a robot). In social science, the manifold is constructed from data via the VAE. But once the manifold is in hand, the analytical tools — Lie derivatives, covariant derivatives, scalar curvature, geodesics, Morse functions — are the same tools of Riemannian geometry, irrespective of the discipline from which the manifold arose.

3. Vector Autoregression, Granger Causality, and the Chow Test: A Geometric Extension

3.1 VAR and the Lie-derivative flow

The VAR model (Hamilton, 1994) represents a k-variate time series y_t as

$$y_t = A_1 y_{t-1} + A_2 y_{t-2} + \dots + A_p y_{t-p} + u_t$$

and captures lagged inter-variable relationships through the coefficient matrices $A_1, \dots, A_p$. The impulse response function (IRF) traces the propagation of an exogenous shock to one variable through the system over time.

On the GI-theoretic manifold, the lag structure of the VAR is reinterpreted as the time evolution of the flow generated by a vector field X on the manifold. The delay with which the effect of a policy X reaches a variable y_k is measured as the geometric distance — along the manifold — that the flow must traverse to reach the region of the manifold corresponding to y_k.

The linearity of the VAR implicitly assumes that the underlying space is flat. The Riemannian manifold of GI theory is, in general, curved; consequently, the lag structure may vary across the manifold. In a region of high curvature (structural instability), shocks propagate differently from the way they propagate in a region of low curvature (structural stability). This positional dependence of the lag structure is invisible to the linear VAR.

3.2 Granger causality and the Lie bracket

The Granger causality test (Granger, 1969) examines whether the past values of a variable x contribute statistically significant information to the prediction of a variable y, and thereby infers the direction of causality. This is, however, predictive causality — temporal precedence — rather than structural causality.

On the manifold, the direction of causality is captured by the Lie bracket [X, Y] = XY - YX of two policy vector fields X and Y. A non-vanishing Lie bracket signifies that executing X before Y yields a different endpoint from executing Y before X. This non-commutativity is the geometric expression of the proposition that the order of policy execution affects the outcome — a notion of structural causality that is more fundamental than the temporal precedence tested by the Granger procedure.

3.3 The Chow test and scalar curvature

The Chow test (Chow, 1960) tests whether the coefficients of a linear regression model differ significantly between two subsamples separated by a candidate break date, and thereby detects the presence of a structural break.

The scalar curvature R(z) of the GI-theoretic manifold captures the total curvature at each point. A sharp change in scalar curvature reflects a transformation in the structural properties of the policy environment. A transition from positive to negative curvature corresponds to a shift from structural stability to structural instability.

Whereas the Chow test determines the presence or absence of a structural break in binary fashion, the scalar curvature provides a continuous description of the shape of the structural change — the geometric meaning of the transition from positive to negative curvature, the velocity of the transition, and the degree of structural stability before and after. The Chow test says "the structure changed at this point"; the scalar curvature says "the structure changed from stability to instability, and the transition was abrupt."

4. The Lucas Critique: A Geometric Answer

4.1 The critique

Lucas (1976) demonstrated that when a policy regime changes, the expectation-formation functions of economic agents change in response, altering the structural parameters of the economy. Consequently, policy simulations based on fixed-parameter structural models (such as the Cowles Commission simultaneous equations models then in widespread use) are systematically invalid.

The Lucas critique exerted a transformative influence on the methodology of macroeconomics. The large-scale macroeconometric models of the preceding era were rendered, in principle, untenable, and were progressively supplanted by dynamic stochastic general equilibrium (DSGE) models. DSGE models derive the behaviour of economic agents from explicit microeconomic optimisation, and thereby seek to capture endogenously the behavioural responses to policy changes.

DSGE models are not, however, assumption-free. They rest upon specific functional forms for utility, production, and expectations formation, and upon the assumption of rational expectations. If these assumptions are misspecified, the DSGE model is itself within the ambit of the Lucas critique.

4.2 Geometric reformulation

In the language of Riemannian geometry, the Lucas critique may be restated as follows.

When the policy vector field X deforms the metric g — that is, when the Lie derivative satisfies $L_X g \neq 0$ — any prediction predicated upon a fixed metric g_0 is invalid.

This is the error of employing a fixed-metric model when the metric is not a Killing field. A Killing field is a vector field along which the metric is preserved: $L_X g = 0$. Change along a Killing field is structure-preserving — superficial. When the metric is not a Killing field, the structure of the space itself is changing, and any analysis that presupposes a fixed structure is, by construction, mistaken.

4.3 Comparison with DSGE

The DSGE programme responds to the Lucas critique by constructing a model in which agents' behavioural responses to policy changes are derived endogenously from optimisation. The strategy is to build a model sufficiently rich that structural change is captured within it.

The Lie derivative of GI theory adopts a fundamentally different strategy. Rather than building a model that captures structural change endogenously, the Lie derivative measures structural change directly from the data, without presupposing any specific model of agent behaviour. The Lie derivative $L_X g$ quantifies the degree to which a policy X deforms the geometric structure of the environment, in a manner that is free of assumptions regarding utility functions, production functions, or the rationality of expectations.

This is, we submit, a response to the Lucas critique that operates on a different plane from that of DSGE. Where DSGE seeks to build the correct model, GI theory seeks to dispense with the need for a structural model altogether — measuring structural change model-free, directly from the geometry of the data.

5. The Synthetic Control Method on the Manifold

The synthetic control method (Abadie and Gardeazabal, 2003; Abadie, Diamond, and Hainmueller, 2010) constructs a counterfactual control unit as a weighted average of untreated units, and estimates the causal effect of an intervention by comparing the treated unit with its synthetic counterpart.

On the GI-theoretic manifold, the counterfactual is realised by constructing a counterfactual manifold. One removes the effect of the treatment from the dataset, retrains the VAE, and obtains a hypothetical manifold (Z', g'). The Lie derivative and scalar curvature of the original and counterfactual manifolds are then compared.

The synthetic control method requires the existence of comparable untreated units from which to construct the synthetic control. The manifold-based approach of GI theory constructs the counterfactual from the treated unit's own data alone, and is therefore applicable in cases where no comparable control group exists — a circumstance that arises frequently in historical and political analysis, where the events under study are singular and without parallel.

6. The Inner/Outer Decomposition of the Metric and the Manifold Embedding of Social Networks

6.1 Decomposition of the metric tensor

One may decompose the metric tensor g into components corresponding to internal and external variables: g = g_internal + g_external. Computing the Lie derivative separately for each component — ||L_X g_internal|| and ||L_X g_external|| — permits the quantitative identification of whether a policy intervention transformed the internal structure of an organisation or state, or its external environment.

This decomposition provides a geometric formalisation of the ubiquitous distinction in social science between internal and external factors — between organisational reform and environmental change, between domestic policy and the international milieu.

6.2 Manifold embedding of social networks

Social network analysis in the tradition of Granovetter (1973) operates on discrete graph structures — nodes and edges — and employs graph-theoretic quantities such as degree centrality, betweenness centrality, and clustering coefficients. GI theory embeds the network into a continuous Riemannian manifold by using node feature vectors as input to the VAE, thereby extending the discrete graph structure to continuous geometry.

This embedding makes available, in addition to the standard network-analytic quantities, the full apparatus of Riemannian geometry: curvature, geodesic distance, and the Lie derivative. Structural changes in the network may be tracked by the Lie derivative, and structural turning points detected by discontinuities in the scalar curvature.

7. Morse Theory and Agent-Based Models on Manifolds

7.1 Morse-theoretic analysis

The critical points of a Morse function (Milnor, 1963) on the manifold — local maxima (peaks), local minima (troughs), and saddle points — classify the topological features of the policy environment. The Morse inequalities relate the number of critical points to the Betti numbers (the topological "holes") of the manifold, quantifying the overall structural complexity of the environment.

7.2 Agent-based models on curved spaces

The agent-based model (ABM) of Epstein and Axtell (1996) simulates the behaviour of many agents acting according to prescribed rules on a spatial grid, from which macroscopic patterns emerge. In existing ABMs, the agents move on a flat space (a lattice or Euclidean plane).

Within the GI-theoretic framework, the behavioural rules of agents may be formulated as vector fields on the manifold, so that agents move on a curved space. Since the curvature of the space affects the outcome of the agents' actions, emergent patterns may arise that are invisible in flat-space ABMs.

8. A Common Language on Manifolds: Towards Cross-Disciplinary Integration

The most consequential observation to emerge from the foregoing discussion is that the method of constructing the manifold is discipline-dependent, but the analytical tools available on the constructed manifold are discipline-independent.

In physics, the manifold is derived from theory: the four-dimensional Lorentzian manifold of general relativity, the configuration space of a robotic manipulator, the rotation group SO(3) of an aircraft's

attitude. In social science, the manifold is constructed from data via the VAE. But once the manifold has been constructed — by whatever means and in whatever discipline — the Lie derivative is the same Lie derivative, the scalar curvature is the same scalar curvature, and the Morse function is the same Morse function. The mathematics is indifferent to the provenance of the manifold.

This suggests that quantitative social scientists may share a common mathematical language with physicists, chemists, and engineers for the analysis of data on manifolds. Just as Fogel and North imported the toolkit of econometrics into the discipline of history, GI theory proposes to import the toolkit of Riemannian geometry into the whole of social science.

9. Honest Limitations and Directions for Further Research

This paper is a methodological proposal, and we wish to be explicit about its limitations.

(1) The precise scope of "model-free." The Lie derivative of GI theory is free of structural model assumptions — it does not presuppose a utility function, a production function, or the rationality of expectations. It is, in this sense, structural-model-free. It is not, however, assumption-free in toto: the construction of the manifold depends on the VAE, and the VAE is itself a model with an architecture and hyperparameters. The precise characterisation is therefore "structural-model-free but generative-model-dependent." This dependence is, we contend, qualitatively different from the structural-model dependence of DSGE: the VAE's architecture can be varied and the robustness of the results verified empirically, whereas the structural assumptions of a DSGE model (rational expectations, specific functional forms) are not easily subjected to the same kind of empirical robustness check.

(2) The character of the correspondences. The correspondences proposed in this paper between existing methods and GI-theoretic concepts (e.g. Granger causality ↔ Lie bracket, Chow test ↔ scalar curvature) are functional analogies, not formal equivalence theorems. Granger causality is a statistical test; the Lie bracket is a differential-geometric operation. We do not claim to have proved that the former is a special case of the latter. We claim, more modestly, that these correspondences are illuminating, that they reveal structural information invisible to the existing methods, and that their rigorous mathematical formalisation is a task for future work.

(3) Empirical validation. This paper is a theoretical and methodological proposal. Large-scale empirical validation is deferred to individual case studies. The application of the GI theory pipeline to real-world social science data (national statistics, financial data, voting records, historical records) is a task for subsequent research. The forthcoming case-study series will demonstrate, on domain-specific datasets, the numerical stability, computational feasibility, and interpretative value of the correspondences proposed herein.

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This paper is a methodological proposal. The correspondences presented herein await rigorous mathematical formalisation and large-scale empirical validation. The author is not affiliated with any institution.